Vector Spaces IV: Quotient Spaces and Counting over Finite Fields

From the class book: Chapter 3, Sections 3.6 and 3.7

1. Key definitions

Cosets. For a subspace W of V, the coset of v is $v + W = \{v + w : w \in W\}$, and $v_1 + W = v_2 + W \iff v_1 - v_2 \in W$. Geometrically, the cosets of a line W in $\mathbb{R}^2$ are the lines parallel to W.

Quotient space. V/W is the set of cosets with $(v_1 + W) + (v_2 + W) = (v_1 + v_2) + W$ and $a(v + W) = av + W$. Its zero is $W = 0 + W$, and the map $v \mapsto v + W$ is linear and onto with kernel W.

- $v + W$ is a subspace of V only when $v \in W$ (then $v + W = W$).
- $\mathbb{F}_q$ is the field with q elements; an n-dimensional space over $\mathbb{F}_q$ has q^n vectors.

2. Main results

Important results: quotient spaces

- $\dim V/W = \dim V - \dim W$ when V is finite dimensional. If $\{w_1, \ldots, w_k\}$ extends to a basis $\{w_1, \ldots, w_k, v_1, \ldots, v_m\}$ of V, then $v_1 + W, \ldots, v_m + W$ is a basis of V/W.
- Any complement W' of W is isomorphic to V/W.
- $(W_1 + W_2)/W_2 \cong W_1/(W_1 \cap W_2)$: both have dimension $\dim(W_1 + W_2) - \dim W_2$.
- If $T : V \to U$ is linear and onto with kernel W, then $V/W \cong U$. Example: $p \mapsto (p(a_1), \ldots, p(a_k))$ shows $P_n/\{p(a_i) = 0 \ \forall i\} \cong \mathbb{R}^k$ for distinct a_i, $k \leq n + 1$.
- If $v_1 + W, \ldots, v_k + W$ are independent in V/W, then $v_1, \ldots, v_k$ are independent in V. The converse fails.
- W finite dimensional and V infinite dimensional: V/W is infinite dimensional. Both infinite: V/W may be either ($\mathbb{R}[x]/\{p(0) = 0\} \cong \mathbb{R}$, but $\mathbb{R}[x]/\{\text{even}\}$ is infinite).

Theorem (Counting over $\mathbb{F}_q$, $\dim V = n$)

- $|V| = q^n$; if $q = p^k$, $|V| = p^{nk}$. For a subspace W of dimension k, $|V/W| = q^{n-k}$.
- Ordered independent k-tuples: $(q^n - 1)(q^n - q)\cdots(q^n - q^{k-1})$.
- Ordered bases $= |GL_n(\mathbb{F}_q)| = (q^n - 1)(q^n - q)\cdots(q^n - q^{n-1})$; unordered bases: divide by $n!$.
- k-dimensional subspaces: the Gaussian binomial $\binom{n}{k}_q = \dfrac{(q^n - 1)(q^n - q)\cdots(q^n - q^{k-1})}{(q^k - 1)(q^k - q)\cdots(q^k - q^{k-1})}$, and $\binom{n}{k}_q = \binom{n}{n-k}_q$.
- Lines (and hyperplanes): $\dfrac{q^n - 1}{q - 1}$.

Space	Lines	Planes	Ordered bases
$\mathbb{F}_2^2$	3	1	6
$\mathbb{F}_3^2$	4	1	48
$\mathbb{F}_2^3$	7	7	168
$\mathbb{F}_3^3$	13	13	11232
$\mathbb{F}_2^4$	15	35	20160
$\mathbb{F}_3^4$	40	130	24261120

3. Methods

Method: counting subspaces or bases over $\mathbb{F}_q$.

- (1) Count ordered independent tuples: the j-th vector avoids the span of the previous ones, giving $q^n - q^{j-1}$ choices.
- (2) For k-dimensional subspaces, divide the number of ordered independent k-tuples in V by the number of ordered bases of one k-dimensional space.
- (3) For lines use $(q^n - 1)/(q - 1)$: each line has $q - 1$ non-zero vectors.

Method: dimension of a quotient.

- Find $\dim W$ (from equations: n— rank), then subtract.
- Or find an onto linear map with kernel W; its target is V/W up to isomorphism.
- To test $u + W = v + W$, check whether $u - v \in W$.

4. Exam shortcuts and common traps

- $\dim M_n(\mathbb{R})/\{\text{symmetric}\} = \frac{n(n-1)}{2}$ and $\dim M_n(\mathbb{R})/\{\text{tr} = 0\} = 1$.
- Total subspaces of $\mathbb{F}_2^3$: $1 + 7 + 7 + 1 = 16$; of $\mathbb{F}_2^2$: $1 + 3 + 1 = 5$.

- In $\mathbb{F}_3^2$, distinct non-zero u, v are dependent iff $v = -u$: exactly 4 such sets, one per line.
- **Traps:** $v + W$ is not a subspace unless $v \in W$; independence in V does not pass to V/W; unordered bases are ordered bases divided by $n!$; $\mathbb{F}_4$ is not $\mathbb{Z}_4$, and $\mathbb{F}_4^2$ has 5 lines.

5. Solved examples

Example 1. Let $W = \operatorname{span}\{(1,2)\} \subset \mathbb{R}^2$. Decide whether $(1, \frac{1}{2}) + W = (6, \frac{21}{2}) + W$, and whether $(3,1) + W = (4,3) + W$.

Solution

- $(1, \frac{1}{2}) - (6, \frac{21}{2}) = (-5, -10) = -5(1,2) \in W$: equal.
- $(3,1) - (4,3) = (-1,-2) \in W$: equal.

Example 2. Find $\dim V/W$ for $V = P_5(\mathbb{R})$, $W = \{p : p(1) = p(2) = p(3) = 0\}$.

Solution

- $T(p) = (p(1), p(2), p(3))$ is onto $\mathbb{R}^3$ (Lagrange interpolation) with kernel W.
- So $V/W \cong \mathbb{R}^3$: dimension 3, and $\dim W = 6 - 3 = 3$ ($W = (x-1)(x-2)(x-3)P_2$).

Example 3. Count the two-dimensional subspaces of $\mathbb{F}_3^4$.

Solution

- Ordered independent pairs in $\mathbb{F}_3^4$: $(81-1)(81-3) = 6240$.
- Ordered bases of one plane: $(9-1)(9-3) = 48$.
- Number of planes: $6240/48 = 130$.

Example 4. Count the unordered bases of $\mathbb{F}_2^3$, and the ordered independent pairs in $\mathbb{F}_2^4$.

Solution

- $|GL_3(\mathbb{F}_2)| = 7 \cdot 6 \cdot 4 = 168$ ordered bases, so $168/3! = 28$ unordered ones.
- Pairs in $\mathbb{F}_2^4$: $15 \cdot 14 = 210$.

Example 5. Let $V = \mathbb{R}[x]$. Find $\dim V/W$ for $W = \{p : p(0) = p'(0) = 0\}$ and for $W =$ even polynomials.

Solution

- $p \mapsto (p(0), p'(0))$ is onto $\mathbb{R}^2$ with kernel W: $\dim V/W = 2$.
- For even polynomials, $V = \{\text{even}\} \oplus \{\text{odd}\}$, so $V/W \cong \{\text{odd}\}$: infinite dimensional.

Example 6. In $\mathbb{R}^6$, $\dim W_1 = 4$, $\dim W_2 = 3$, $\dim(W_1 \cap W_2) = 2$. Find $\dim(W_1 + W_2)/W_2$, $\dim W_1/(W_1 \cap W_2)$ and $\dim \mathbb{R}^6/(W_1 + W_2)$.

Solution

- $\dim(W_1 + W_2) = 4 + 3 - 2 = 5$.
- $\dim(W_1 + W_2)/W_2 = 5 - 3 = 2 = \dim W_1/(W_1 \cap W_2) = 4 - 2$, as the isomorphism theorem says.
- $\dim \mathbb{R}^6/(W_1 + W_2) = 6 - 5 = 1$.

Example 7. How many elements does V/W have, for $V = \mathbb{F}_3^5$ and $\dim W = 2$? How many cosets of W contain a given vector?

Solution

- V/W is 3-dimensional over $\mathbb{F}_3$: $3^3 = 27$ cosets, each with $|W| = 9$ vectors.
- The cosets partition V, so each vector lies in exactly one coset.